Heltec V3 vs Heltec V4 vs NodeMCU ESP32

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Short answer: the Heltec V4 is the best all-rounder and the clear pick for long-range LoRa and Meshtastic. The Heltec V3 does the same jobs with less power and memory. The NodeMCU ESP32 has no LoRa or screen, but it's the only one of the three with Bluetooth Classic audio and real analog DAC outputs, and it has the most free pins.

All three are ESP32 boards, but the two Heltecs use the newer **ESP32-S3** chip and add an SX1262 LoRa radio, a 0.96″ OLED and battery charging. The NodeMCU uses the **classic ESP32** and is just the bare chip with USB.

Side-by-side specs

Feature	Heltec WiFi LoRa 32 V3	Heltec WiFi LoRa 32 V4	NodeMCU ESP32 (DevKit V1)
Chip	ESP32-S3FN8	ESP32-S3R2 (V4-R8: ESP32-S3R8)	ESP32-D0WD (classic) in WROOM-32
CPU	2× Xtensa LX7, 240 MHz	2× Xtensa LX7, 240 MHz	2× Xtensa LX6, 240 MHz
SRAM	512 KB	512 KB	520 KB
PSRAM	None	2 MB (8 MB on V4-R8)	None
Flash	8 MB (in chip)	16 MB (external)	4 MB typical
Wi-Fi	802.11 b/g/n, 2.4 GHz	802.11 b/g/n, 2.4 GHz	802.11 b/g/n, 2.4 GHz
Bluetooth	BLE 5	BLE 5	Bluetooth 4.2 Classic + BLE
LoRa radio	SX1262	SX1262 + power amplifier	None
LoRa bands	433, 470–510, 863–870, 902–928 MHz	863–928 MHz (high power); 433–510 MHz low-power version	—

Feature	Heltec WiFi LoRa 32 V3	Heltec WiFi LoRa 32 V4	NodeMCU ESP32 (DevKit V1)
LoRa max TX power	21 ± 1 dBm (~125 mW)	28 ± 1 dBm (~630 mW), 22 dBm low-power version	—
LoRa RX sensitivity	about −134 to −137 dBm	−137 dBm (SF12, 125 kHz)	—
Display	0.96″ 128×64 OLED	0.96″ 128×64 OLED in a PC protective frame	None
Wi-Fi antenna	Metal spring antenna	FPC antenna + IPEX connector for an external one	PCB antenna on module
USB	USB-C via CP2102 bridge	USB-C, native ESP32-S3 USB (CP2102 removed)	Micro-USB (some USB-C) via CP2102 or CH340
Battery	Li-ion charging, SH1.25 2-pin	Improved Li-ion charging, SH1.25 2-pin	None built in
Solar input	No	Yes, SH1.25 2-pin	No
GPS/GNSS port	No	Yes, SH1.25 8-pin	No
DAC (analog out)	No	No	2 channels (GPIO 25, 26)
Touch pins	Yes (S3 has 14)	Yes (S3 has 14)	10
Deep-sleep current (board)	Low tens of µA	Under 20 µA	Several mA (regulator, USB chip, LED)
Size	51.7 × 25.4 mm	51.7 × 25.4 × 10.7 mm	about 52 × 28 mm
Price (Nov 2026)	about $18–20	about $27	about $5–10

The V4's 28 dBm is about 4–5× the V3's radio power, which in practice means noticeably longer Meshtastic hops. Check your local rules: in the US, 902–928 MHz LoRa nodes run unlicensed under FCC Part 15, and your ham license allows more on the 33 cm band with your callsign (Meshtastic's licensed mode turns off encryption).

ESP32-S3 vs classic ESP32: what each chip can do

The S3 in the Heltecs is newer and faster at math, but the classic chip in the NodeMCU keeps a few features the S3 dropped.

Capability	ESP32-S3 (Heltec V3, V4)	Classic ESP32 (NodeMCU)
CPU core	LX7, with vector (SIMD) instructions for AI and DSP	LX6
Native USB	Yes: USB serial/JTAG and USB OTG (can act as a keyboard, mouse, MIDI or flash drive)	No, needs a USB-serial chip
Bluetooth Classic	No	**Yes**: A2DP audio, SPP serial, old gamepads
Bluetooth LE	BLE 5, long-range coded PHY, 2 Mbps	BLE 4.2
DAC (true analog out)	No	**Yes**, 2 channels
GPIOs on the chip	45	34
ADC channels	20	18
Touch inputs	14	10
Camera / LCD interface	Yes (parallel LCD and camera)	Via I²S tricks only
Ethernet MAC	No (SPI Ethernet chips work)	**Yes** (RMII)
PSRAM support	Fast octal PSRAM (V4 has it built in)	Quad PSRAM (WROVER modules only)
ADC2 + Wi-Fi conflict	Mostly fixed	ADC2 unusable while Wi-Fi runs

What that means for you:

- Want a Bluetooth speaker, an A2DP audio player or a PS3/PS4 controller link? Only the **NodeMCU** can do it.

- Want 8-bit analog audio or composite video straight off a pin? Only the **NodeMCU** has DACs (Heltecs can still do audio through I²S or PWM).
- Want the board to show up as a USB keyboard, mouse or MIDI device? The **V4** is easiest, thanks to native USB. The V3's USB-C port goes through a CP2102, so its native USB pins aren't wired to the jack.
- Heavy screens, web pages or lots of buffers? The **V4's PSRAM** gives you megabytes instead of hundreds of kilobytes.

What changed from V3 to V4

The V4 keeps the V3's size and pinout, so V3 cases, shields and most code carry over, but nearly every part around the chip was upgraded.

Area	V3	V4	Why it matters
LoRa power	21 dBm	28 dBm with a power amplifier (FEM)	Longer range
Memory	8 MB flash, no PSRAM	16 MB flash + 2 MB PSRAM (8 MB on V4-R8)	Room for big firmware, UI, Wi-Fi + BLE together
USB	CP2102 bridge	Native S3 USB	No driver needed; USB device modes possible
Screen	Bare OLED	OLED in a protective PC frame	Less cracked screens
Wi-Fi antenna	Spring antenna	FPC antenna + IPEX jack	Better Wi-Fi range, external antenna option
Power	Li-ion charger	Better charger + solar input	Solar-powered mesh nodes
GNSS	None	8-pin GPS connector	Plug-in GPS for tracking
Sleep current	Higher	Under 20 µA	Longer battery life

Firmware and pin gotchas on the V4

- **Pick the right firmware.** Meshtastic has separate builds: `firmware-heltec-v4` for the normal V4 and `firmware-heltec-v4-r8` for the R8 version. V3 firmware is not the same build.

- **Select the LoRa amplifier in Arduino.** Heltec's board package needs the right FEM option: `USE_GC1109_PA` for V4.2 boards and `USE_KCT8103L_PA` for V4.3 boards.
- **The V4-R8 moves some control pins.** On the standard V4: Vext control GPIO 36, GNSS power GPIO 34, LED GPIO 35, GNSS reset GPIO 42. On the V4-R8: Vext GPIO 40, GNSS power GPIO 42, LED GPIO 46, and no GNSS reset pin. Your ESP32 DJ, MP3, Scout and Shelf builds should check which one they're on.

Pins: what's taken and what's free

The Heltecs spend about a dozen pins on the LoRa radio, screen, button and power control, while the NodeMCU leaves everything except the flash and USB-serial pins to you.

Job	Heltec V3	Heltec V4	NodeMCU ESP32
LoRa SPI (NSS, SCK, MOSI, MISO)	GPIO 8, 9, 10, 11	GPIO 8, 9, 10, 11	—
LoRa RST, BUSY, DIO1	GPIO 12, 13, 14	GPIO 12, 13, 14	—
LoRa amplifier control	—	Extra PA/LNA pins (vary by revision)	—
OLED I²C (SDA, SCL, RST)	GPIO 17, 18, 21	GPIO 17, 18, 21	—
User button (PRG/BOOT)	GPIO 0	GPIO 0	GPIO 0 (BOOT, not on 30-pin header)
Onboard LED	GPIO 35	GPIO 35 (R8: 46)	GPIO 2
Vext power switch for OLED/sensors	GPIO 36	GPIO 36 (R8: 40)	—
Battery voltage read	GPIO 1 (enable on 37)	GPIO 1	—
GNSS power / reset	—	GPIO 34 / 42 (R8: 42 / none)	—
USB	via CP2102 on UART0 (43, 44)	Native USB on GPIO 19, 20	via CP2102/CH340 on UART0 (1, 3)

Job	Heltec V3	Heltec V4	NodeMCU ESP32
Flash / PSRAM (never touch)	GPIO 26–32	GPIO 26–32 (R8 also 33–37)	GPIO 6–11
Strapping pins (careful at boot)	0, 3, 45, 46	0, 3, 45, 46	0, 2, 5, 12, 15
Pins left for your projects	About 15–18	About 14–16	About 22 (14 with no catches)

Good free pins on both Heltecs are usually GPIO 2–7 (also analog inputs), 38–42 and 47–48. The V3 also frees GPIO 19–20, which the V4 uses for USB. Meshtastic uses GPIO 41/42 for an extra I²C sensor bus. Check Heltec's pin map for your exact revision before wiring.

The NodeMCU's advantage: no radio or screen taking pins, two DACs, and the GPIO matrix routes UART, SPI, I²C or PWM to nearly any of its pins. See the ESP32 Classic Field Guide for the full pinout.

Which board for which project

Project	Best board	Why
Meshtastic / MeshCore node, reaching far away	Heltec V4	28 dBm radio, better sensitivity, solar input
Solar relay node on a pole or roof	Heltec V4	Solar connector, low sleep current
GPS tracker	Heltec V4	Built-in GNSS port
Cheap extra mesh node or handheld	Heltec V3	Same LoRa features, lower price
LoRa sensor that sends a reading every few minutes	Heltec V3 or V4	Both sleep well on a battery
FT8 / PSK31 over LoRa hardware (LoFT8, LoPSK)	Heltec V4	More RAM and flash for decoding
Emulators with web view (C64, Mac, Apple II)	Heltec V4	PSRAM makes room for frame buffers
Web server / file hub (Shelf Lite)	Heltec V4	16 MB flash, PSRAM

Project	Best board	Why
Bluetooth speaker or A2DP music player	NodeMCU	Only board with Bluetooth Classic
VGA or composite video computer (TidyOS)	NodeMCU	DAC and lots of free pins
Lots of sensors, relays and LEDs	NodeMCU	Most free pins, no radio or screen in the way
USB keyboard / MIDI gadget	Heltec V4	Native USB on the jack
Cheapest Wi-Fi project	NodeMCU	About $5–10

Bottom line: buy V4s for anything LoRa from now on, keep your V3s as extra mesh nodes and test boards, and use the NodeMCU for Bluetooth audio, video tricks and pin-hungry builds.

Sources

- Heltec WiFi LoRa 32 V4 product page
- Heltec WiFi LoRa 32 V4 wiki
- WiFi LoRa 32 V4.3.1 datasheet
- Heltec V4 and V4-R8 pin differences
- Heltec WiFi LoRa 32 V3 product page
- Rokland V3 vs V4 comparison
- Meshtastic: Heltec LoRa 32 devices
- ESPHome: Heltec WiFi LoRa 32 V3 pins
- Espressif ESP32 Series Datasheet

Prices are typical US retail in November 2026. ESP32-S3 chip feature counts (GPIOs, ADC, touch, USB) and the Heltec pin roles not listed on the pages above are from Espressif's S3 documentation and common board pin maps; double-check against the datasheet for your board revision.